Track 01: Synthetic Eye Movement & Virtual Camera Rendering

XR Hackathon

Team Name : **MarkovX**

College Name : **SRM Institute of Science and Technology**

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THE PROBLEM

CONTEXT

Gaze estimators for XR headsets need very large sets of eye images with exact gaze labels.

Real capture at extreme angles is slow, costly and privacy-restricted.

Our approach: a calibrated photoreal digital head that renders unlimited labelled frames.

Both Samsung specs are met: the 10-frame calibration task and the textures / videos / .blend deliverable.

REQUIREMENTS WE MEET

Item	Spec
Eye motion	3D rotation about eyeball centre
Fixation target	35 cm, +-5 cm
Camera	40 cm, locked
Render	1024 x 1024 PNG
Clockwise set	10 frames, theta = 2*pi*k/10
Inputs	h = sin(theta), v = cos(theta)
Ground truth	metadata.json per frame
Check	Left / Right / Up / Down / Reset

SOLUTION

Head

Google GNM Head v3 (17,821 vertices), converted to a metric world frame, eye midpoint at the origin.

Eye rig

One bone per eyeball at its centre, tracking a shared fixation target: shortest-arc rotation, natural vergence of about 5.1 degrees per eye.

Lids, brows, pupils

Shape keys hinge the lids about the eyeball, so no cornea intersection. Lid-follow drivers use simple expressions only.

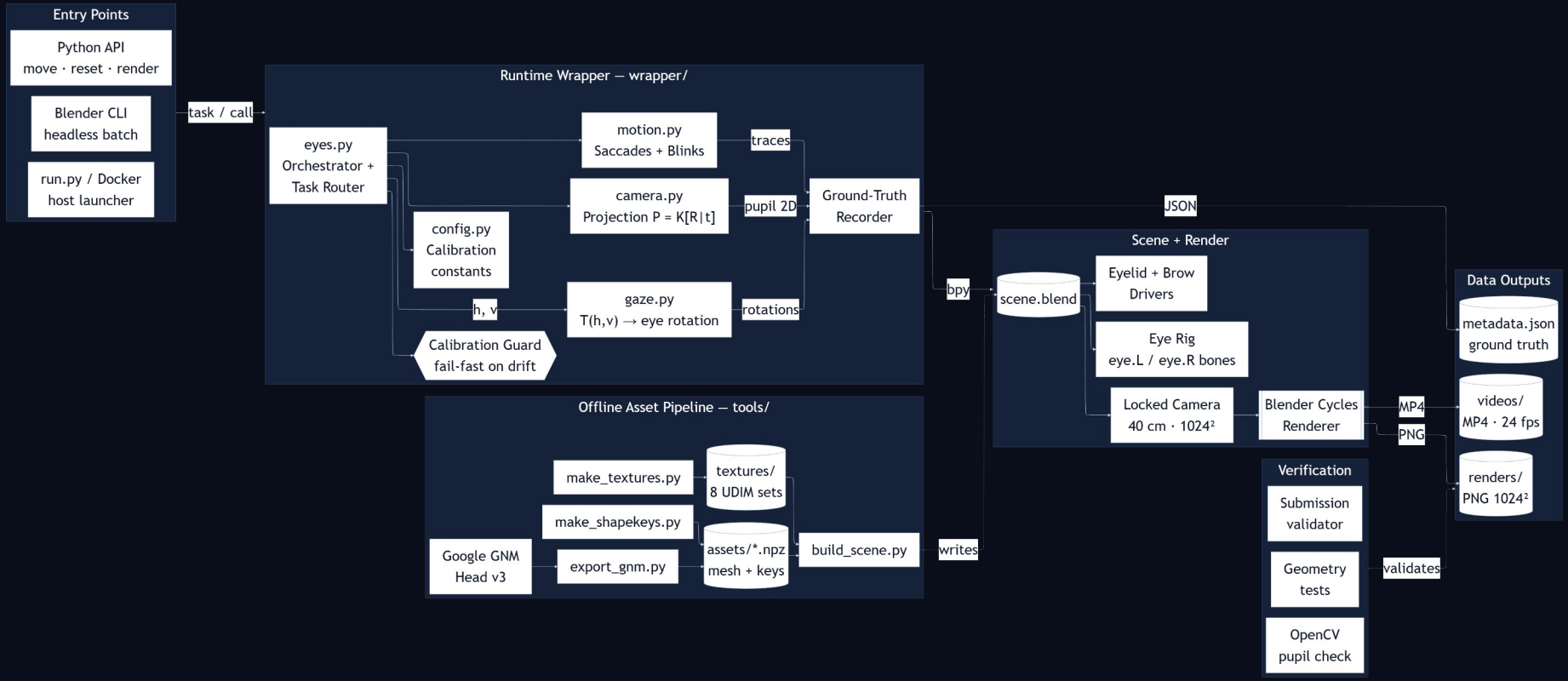
Look

8 procedural skin sets in GNM UDIM layout with different left/right irises. Cycles with OIDN denoise, constant lighting.

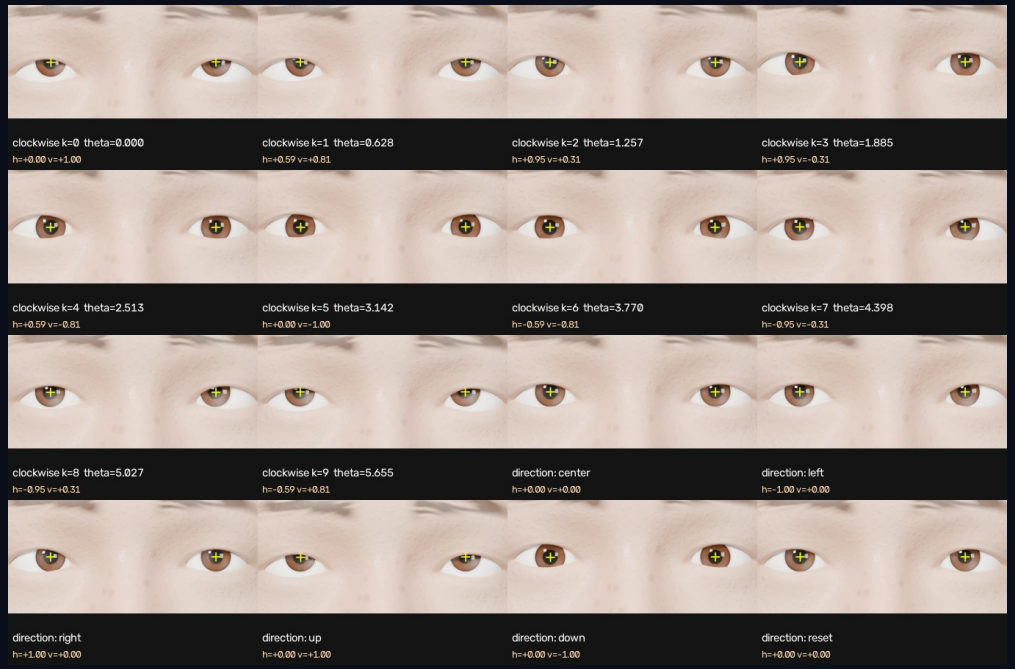


Portrait render, texture set 01

ARCHITECTURE



Track Final Output



Clockwise k=0..9 and Left / Right / Up / Down / Reset. Crosses mark metadata pupil centres.

```
{ "frame": 3,
  "theta": 1.885,
  "input": {"h": 0.951, "v": -0.309},
  "eye_rotation_L": [rx, ry, rz],
  "eye_rotation_R": [rx, ry, rz],
  "pupil_center_2D_L": [x, y],
  "pupil_center_2D_R": [x, y],
  "camera_matrix": 3x4 }
```



8 texture sets

3 videos, 12 s

10 + 6 renders

DIFFERENTIATION

Aspect	Typical approach	Ours
Eye motion	Parallel eyes or 2D pupil slides	True 3D rotation, both eyes converge on one 35 cm target
Verification	Visual check	Independent OpenCV pupil detector, 5 / 5 directions
Calibration	Manual, drifts	Locked and checked at runtime; refuses to render if violated
Eyelids	Static or clipping	Hinge about the eyeball, follow gaze through drivers
Motion	Linear tweens	Min-jerk saccades, blinks, microsaccades, eye roll
Textures	Single skin	8 sets in GNM UDIM layout, distinct L/R irises

DEMO & LOCAL SETUP

RUN IT (Blender 4.5 LTS)

```
$ blender -b scene.blend -P wrapper/eyes.py -- clockwise
$ blender -b scene.blend -P wrapper/eyes.py -- directions
$ python run.py --input left | right | generate
$ docker build -t eyes . && docker run --rm eyes --clockwise
```

PYTHON API

```
import eyes
eyes.move(h, v)    # h, v in [-1, 1]
eyes.reset()       # centre, converge on target
eyes.render('out.png')
```



Videos: Left, Right, Generate Motion, 12 s each

Repo: github.com/AzamAliCodes/synthetic-eye-movement-vcr Demo video: videos/demo_video.mp4

Check	Result
Direction check (OpenCV)	5 / 5 pass
Clockwise order	pass
Pupil centre error	mean 4.5 px, max 7.9 px
Camera / target distance	0.40 m / 0.35 m
Image size	1024 x 1024, 10 frames
Submission validator	25 / 25
Videos	12 s each (limit 15 s)

LIMITS

CPU Cycles: about 1 to 2 min per still

Procedural skin, bald head, painted brows

About 5 px offset between geometric and detected pupil (corneal refraction)

NEXT

Random identities, lighting and glasses

Full gaze datasets with labels

Thank You